

Exercise 1

The exercise is to investigate stability properties of the origin for the nonlinear system

$$x(k+1) = \underbrace{\begin{pmatrix} x_1(k) - x_1(k)^3 \\ -x_2(k)^3 - x_1(k)^3 \end{pmatrix}}_{:=g(x)}, \quad x(0) = x_0. \quad (1)$$

Determine a compact set $0 \subset \Omega \subset \mathbb{R}^2$ such that the origin $x_s = 0$ is a locally asymptotically stable equilibrium point for system (1) for all $x \in \Omega$.

Exercise 2

Consider the following MPC problem

$$\begin{aligned} \min_{u_i} \quad & \alpha x_N^2 + \sum_{i=0}^{N-1} x_i^2 + u_i^2 \\ \text{s.t.} \quad & x_{i+1} = \frac{3}{2}x_i + u_i, \quad i = 0, \dots, N-1 \\ & x_0 = x(k) \end{aligned}$$

with $\alpha \in \mathbb{R}$, and $N = 1$. Is the origin an asymptotically stable equilibrium point for the closed-loop system under the resulting MPC control law for:

- i) $\alpha = \frac{1}{2}$?
- ii) $\alpha = \frac{3}{4}$?